Species-Aware Review: Pyrroloquinoline Quinone (PQQ) Biosynthesis in *Pseudomonas putida* KT2440

Taxon: *Pseudomonas putida* KT2440 (PSEPK, NCBI:txid160488, proteome UP000000556) **Module / bucket:** UniPathway UPA00539 (pyrroloquinoline quinone biosynthesis) **Date:** 2026-07-21

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1. Executive summary

The UPA00539 PQQ-biosynthesis module is **fully satisfiable** in *P. putida* KT2440. All five conserved catalytic steps map to a compact chromosomal cluster, **PP_0376–PP_0381** (`pqqE-pqqD1-pqqC-pqqB-pqqA-pqqF`, i.e. the *pqqFABCDE* operon read 5'→3'): precursor-peptide supply (PqqA), PqqD-assisted radical-SAM cross-linking (PqqD1 + PqqE), PqqB oxygenation, PqqF proteolysis, and PqqC terminal ring closure. PQQ production is functionally confirmed — KT2440 depends on PQQ as the cofactor for its periplasmic glucose dehydrogenase (Gcd) (P 27287323).

Two curation caveats dominate: 1. **Proteolysis route.** KT2440 encodes a **single-chain M16 PqqF** (PP_0381, 766 aa, InterPro IPR011844). The generic module's "two-component PqqF/PqqG M16B protease" alternative derives from the α-proteobacterium *Methylobacterium extorquens* (P 31427437) and **does not apply** here. A gene sometimes labelled *pqqG* in KT2440 (**PP_0375**) is a **Peptidase S9 / prolyl-oligopeptidase**, not an M16B partner — a naming collision that should not be curated as the two-component route. 2. **Paralog ambiguity. PqqD2 (PP_2681)** is a standalone PqqD-family paralog embedded among quinoprotein dehydrogenase genes; its assignment to UPA00539 is **candidate_uncertain** and may be over-propagated. The operon-embedded **PqqD1 (PP_0377)** is the high-confidence chaperone.

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2. Target-organism pathway definition

Included process. Conversion of the ribosomally synthesized precursor peptide **PqqA** into the mature small-molecule redox cofactor **PQQ**. This is a RiPP (ribosomally synthesized and post-translationally modified peptide) pathway comprising: (i) a C–C cross-link between a specific Glu and Tyr in PqqA by the radical-SAM enzyme PqqE, presented by the PqqD RiPP-recognition element (RRE); (ii) proteolytic excision of the cross-linked di-amino-acid intermediate; (iii) PqqB-dependent oxygenation/hydroxylation; and (iv) an eight-electron/eight-proton oxidative ring closure by PqqC yielding PQQ (P 32731194).

Keep separate (neighboring processes, downstream or unrelated). - **PQQ *utilization*** by quinoprotein dehydrogenases — glucose dehydrogenase *gcd* (PP_1444), PQQ-dependent alcohol dehydrogenases (e.g. *pedH/exaA*-type) — is downstream and should not be folded into UPA00539. - **PQQ export/secretion** and the **PedS2/PedR2** and **AgmR** regulatory systems (PMIDs 30158283, 11954793) are regulatory/transport, not biosynthetic. - Adjacent operon genes **PP_0382 (davA)** and **PP_0383 (davB, tryptophan 2-monooxygenase)** belong to lysine/amino-acid catabolism, not PQQ synthesis, despite chromosomal proximity. - Broad overview maps ("cofactor biosynthesis", "metabolic pathways") should not substitute for the specific module.

Alternate names / DB definitions. UniPathway UPA00539; KEGG module for PQQ biosynthesis; "coenzyme PQQ synthesis" (UniProt Pqq nomenclature, letters A–F). Enzyme names: PqqE = "PqqA peptide cyclase" (EC 1.21.98.4); PqqC = "pyrroloquinoline-quinone synthase" (EC 1.3.3.11).

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3. Expected step model (generic → KT2440 status)

#	Generic step	Enzyme/player	KT2440 gene	Status
1	PqqA precursor supply	PqqA precursor peptide	**PP_0380 pqqA**	**covered**
2	PqqD-assisted PqqE cross-link	PqqD RRE + PqqE radical-SAM	**PP_0377 pqqD1** + **PP_0376 pqqE**	**covered**
3	Proteolytic release of intermediate	protease (M16 single-chain *or* M16B two-component *or* other)	**PP_0381 pqqF (single-chain M16)**	**covered (single-chain route)**
4	PqqB-dependent oxygenation	PqqB dual hydroxylase	**PP_0379 pqqB**	**covered**
5	Terminal oxidative ring closure	PqqC synthase	**PP_0378 pqqC**	**covered**

The two alternative proteolysis sub-versions not used here (M16B two-component; generic cellular protease) should be marked **not_expected_in_target_taxon** for KT2440.

4. Candidate genes and evidence

Gene	Locus	Acc	Length	Role / signature	Evidence class	Curation r
pqqA	PP_0380	Q88QV4	23 aa	Precursor peptide; PqqA family (Pfam PF08042). Sequence `MWTKPAYTDLRIGFEVTMYFANR` carries the conserved **E15-x-x-x-Y19** cross-link motif	Homology + motif; operon context; strong	High confidence Short ORF verify it is retained in gene mode
pqqD1	PP_0377	Q88QV7	91 aa	PqqD/RRE chaperone that presents PqqA to PqqE	Homology; strong structural precedent (PMIDs 28481092, 38435514)	High-confidence operon chaperone
pqqE	PP_0376	Q88QV8	376 aa	Radical-SAM PqqA peptide cyclase, EC 1.21.98.4; forms the defining C–C cross-link	Homology to well-characterized PqqE (P 38435514); strong	High confidence **promote t fetch-gene** the module defining catalytic st
pqqB	PP_0379	Q88QV5	303 aa	Metallo-hydrolase/β-lactamase fold (InterPro IPR001279) dual hydroxylase/oxygenase	Homology (P 32731194); strong for step, but **UniProt annotation is outdated**	High confidence structurall but UniPro FUNCTION reads "may involved in transport o PQQ... to th periplasm" (keyword *Transport*) **legacy annotatio** supersede the oxygen role. Upda O2- incorporat

Gene	Locus	Acc	Length	Role / signature	Evidence class	Curation r
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pqqC	PP_0378	Q88QV6	251 aa	Pyrroloquinoline-quinone synthase, EC 1.3.3.11; terminal $8e^{-}/8H^{+}$ ring closure	Homology; well-defined EC; strong	High confidence cofactor-independe oxidase.
pqqF	PP_0381	Q88QV3	766 aa	Single-chain M16 peptidase; InterPro IPR011844 (PQQ_synth_PqqF) + fused C-terminal domains; EC 3.4.24.- (by similarity)	Homology + dedicated PqqF signature; KT2440 expression tracks PQQ (P 27287323); moderate-strong	Broad EC 3.4.24.-; ex cleavage in KT2440 no experimen demonstra **Promote t fetch-gene**
pqqD2	PP_2681	Q88JG8	90 aa	Second PqqD paralog; located far from operon, flanked by *aldB-II* (PP_2680) and *yiaY* alcohol DH (PP_2682)	Homology only; **candidate_uncertain**	UniProt FUNCTION SIMILARIT text and InterPro se **byte-ident to PqqD1** unmodifie similarity i.e. over-propagatio not PqqD2 specific evidence. Likely quinoprote associated accessory. **Promote t fetch-gene** decide.

Direct target-strain evidence: An & Moe (P 27287323) mapped the *pqqFABCDEG* operon in KT2440, showed ≥2 independent transcripts, and found *pqqF* and *pqqB* transcript levels mirror PQQ output — the strongest species-specific data. A 2026 study (P 41705821, Puiggené & Nikel) addresses the "functional organization and regulatory logic" of this cluster and should be consulted directly for updated wiring (abstract not yet indexed).

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5. Gaps, ambiguities, and likely over-annotations

1. **"pqqG" = PP_0375 (Q88QV9) is an S9 prolyl-oligopeptidase, not an M16B PqqG.** Historic operon nomenclature (*pqqFABCDEG*, An & Moe) attached the *pqqG* label to PP_0375, but its family (Pfam PF00326 Peptidase_S9; InterPro IPR001375) is unrelated to the *Methylobacterium* M16B PqqG. **Do not** curate KT2440 as using a two-component PqqF/PqqG protease. PP_0375's role in PQQ synthesis is unproven; treat as accessory/uncertain and, if anything, a candidate for the "alternative cellular-protease" sub-step rather than the M16B route.
2. **PqqD2 (PP_2681) over-propagation risk.** Two PqqD paralogs are annotated to the module; only PqqD1 is operon-embedded. PqqD2's genomic neighborhood argues for a quinoprotein-linked role. Mark candidate_uncertain.
3. **Broad/loose mappings.** PqqF EC 3.4.24.- is family-level (not reaction-specific); PqqE/PqqC have precise ECs. PqqB carries no EC and, more importantly, an **outdated UniProt FUNCTION** ("transport of PQQ... to the periplasm", keyword *Transport*) that predates its reclassification as a dual hydroxylase/oxygenase (metallo-β-lactamase fold, IPR001279; P 32731194). Its GO should be O2-incorporating oxidoreductase, and the *Transport* keyword removed. The historical "PqqB = PQQ transporter" view should not be propagated.
4. **PqqD2 = unmodified copy of PqqD1.** PqqD2 (PP_2681) inherits byte-identical by-similarity FUNCTION/SIMILARITY text and the same InterPro triad as PqqD1 — direct evidence that its UPA00539 membership is annotation transfer, not paralog-specific data.

5. **Order of proteolysis vs. oxygenation** (PqqF vs PqqB) remains biochemically unresolved even in model organisms (P 32731194); this is a pathway-biology gap, not a KT2440 gene gap.
6. **Species transfer.** Mechanistic detail (HDX-MS ternary complexes, PqqD NMR, two-component protease) comes from *Methylobacterium extorquens* and in-vitro systems — **strong for conserved catalytic logic**, **weak for the protease architecture** (α- vs γ-proteobacterial divergence). KT2440-specific evidence is essentially the operon mapping/expression study plus functional PQQ dependence.

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6. Module and GO-curation recommendations

Step	Recommended status
PqqA precursor supply	**covered** (PP_0380)
PqqD-assisted PqqE cross-link	**covered** (PP_0377 + PP_0376)
Proteolysis — single-chain PqqF/M16 route	**covered** (PP_0381)
Proteolysis — two-component PqqF/PqqG M16B route	**not_expected_in_target_taxon**
Proteolysis — alternative cellular protease	**gap/uncertain** (only PP_0375 S9 peptidase as a weak, unproven candidate)
PqqB oxygenation	**covered** (PP_0379)
PqqC terminal ring closure	**covered** (PP_0378)

Module boundary edits. - Constrain the proteolysis step in the KT2440 module instance to the single-chain PqqF alternative; suppress the M16B two-component alternative for this taxon. - Move PqqD2 (PP_2681) to a **candidate_uncertain** / **accessory** slot, not a required core step. - Explicitly exclude PQQ-utilizing quinoprotein dehydrogenases, transport, and the PedS2/PedR2–AgmR regulators from the biosynthetic module.

Annotation-quality fixes (in addition to boundaries). - PqqB (PP_0379): replace the legacy "PQQ transport to periplasm" FUNCTION and remove the *Transport* keyword; annotate as O2-incorporating oxidoreductase/hydroxylase (metallo-β-lactamase fold, IPR001279). Step stays **covered**. - **PqqD2 (PP_2681):** demote from primary UPA00539 member to candidate_uncertain; its annotation is an unmodified copy of PqqD1.

GO term needs. Existing terms suffice: PqqE GO:0018189-type peptide-cyclase (EC 1.21.98.4); PqqC GO:0033735 (pyrroloquinoline-quinone synthase, EC 1.3.3.11); PqqB GO oxidoreductase (paired donors, O2 incorporation). No new GO request appears necessary; if PP_0375 is investigated, a dedicated peptidase term already exists. Recommend a curator note distinguishing "PqqG (S9, KT2440)" from "PqqG (M16B, *Methylobacterium*)" to prevent future over-propagation.

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7. Genes to promote to full `fetch-gene` review

1. **pqqE / PP_0376** — module-defining radical-SAM step; confirm SPASM/radical-SAM Cys motifs and Fe-S cluster residues.
2. **pqqF / PP_0381** — proteolysis assignment; confirm single-chain M16 architecture, catalytic HxxEH Zn motif, and that no M16B partner is required.
3. **pqqD2 / PP_2681** — resolve whether it is a bona fide PQQ-biosynthesis chaperone or a quinoprotein-associated accessory (decides candidate_uncertain vs remove).
4. **PP_0375 (Q88QV9, "pqqG")** — clarify family (S9 peptidase) and whether it has any genuine PQQ-pathway role; prevent M16B mis-annotation.

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8. Key references

- An R, Moe LA. *Regulation of PQQ-dependent glucose dehydrogenase activity in P. putida KT2440.* Appl Environ Microbiol. 2016.

  P 27287323

  — direct KT2440 operon (pqqFABCDEG) mapping and expression.
- Zhu W, Klinman JP. *Biogenesis of the peptide-derived redox cofactor pyrroloquinoline quinone.* 2020.

P 32731194

— *mechanistic review of PqqE, PqqF/G, PqqB.*

- Martins AM, et al. *A two-component protease in Methylobacterium extorquens* (M16B PqqF/PqqG). 2019.

P 31427437

— *establishes α-proteobacterial two-component route (weak transfer to KT2440).*

- Zhu W, Iavarone AT, Klinman JP. *HDX-MS of the multicomponent radical-SAM enzyme PqqE (PqqA/PqqD/PqqE/SAM complexes).* 2024.

P 38435514

- Evans RL, et al. *NMR structure and binding studies of PqqD (RRE).* 2017.

P 28481092

- Puiggené O, Nikel PI. *Functional organization and regulatory logic of the [pqq cluster].* 2026.

P 41705821

— *consult directly for updated KT2440 wiring.*

- Wehrmann M, et al. *PedS2/PedR2 two-component system, rare-earth-element switch, P. putida KT2440.* 2018.

P 30158283

— *regulatory, downstream.*

- Vrionis HC, et al. *AgmR regulator and alcohol utilization in P. putida.* 2002.

P 11954793

— *regulatory, downstream.*

Uncertainty flags: PqqF cleavage specificity and PP_0375/PqqD2 roles are inferred from homology/genomic context, not KT2440 biochemistry. Core catalytic assignments (PqqA/D1/E/B/C) are high-confidence homology with functional PQQ-dependence confirmed in-strain.
